

ITERATION REQUIREMENTS

connection with nodemcu

<https://mytectutor.com/how-to-connect-esp8266-to-mysql-database-using-php-and-arduino-ide/>

Detect spoiled food item (chicken)

- Food safety (material poisoning)
- usability and reusability
- Integration with smartphone
- Accuracy and precision
- Compliance with regulation

FACTORS TO CONSIDER FOR SENSOR

- pH level
- nh3 ammonia
- biogenic amines
- Hydrogen Sulphide
- Water content
- Carbon

DESIGN PHASE

1. **Selection of gases:** We will focus on ammonia and other gases that are emitted by spoiling meat (chicken in this case). Research will be conducted to identify these gases precisely.
2. **Sensor placement:** The meat will be placed in a plastic box equipped with glass sensors. These sensors will be designed to detect the gases emitted by the meat.
3. **Ph Sensor:** We manually use the probe to collect PH data. The probe is connected to the box like an attached pen
4. **pH detection:** A probe (similar to a soil pH detector) will be used to measure the pH level of the meat. This will provide us with another parameter to determine the freshness of the meat.
5. **Data comparison:** The values obtained from the gas sensors and the pH detector will be compared to a dataset of normal pH and gas values. This dataset will be collected from fresh meat in controlled conditions for accuracy.

6. **Output:** If the measured values deviate significantly from the normal range, a signal will be sent to a website or mobile application, indicating that the meat is spoiled. The user interface will be designed for easy interpretation of the results.

DEVELOP PHASE

In the development phase, we will need to purchase several components.

1. **Gas Sensors:** We would require gas sensors that can detect the gases emitted by spoiling meat, especially ammonia. There are several sensors available on the market, such as the MQ-137 sensor, which is specifically designed to detect ammonia levels.
2. **pH Detector:** We would also need a pH detector or probe to measure the pH level of the meat. There are a variety of pH detectors available on the market, with varying degrees of precision and cost.
3. **Microcontroller:** To process the data from the sensors, we would need a microcontroller. An Arduino or Raspberry Pi would be suitable for this purpose.
4. **Other Components:** Depending on the design, we may also need additional components such as a Wi-Fi module for connectivity, a power supply, and casing for the device.

These components can be purchased from various online electronics retailers. In India, sites like [Robu.in](https://www.robu.in/) and [Amazon.in](https://www.amazon.in/) offer a wide range of electronics components.

Once all the parts are acquired, they can be assembled following the design specifications. Remember to test each component individually before assembling the device to ensure they are working correctly.

After assembly, we will need to develop the software that will collect and analyze the data from the sensors, and send notifications if the meat is spoiled. This can be done using various programming languages, depending on the microcontroller used.

Remember to conduct thorough testing of the device in controlled conditions to ensure accuracy and reliability.

CONVERSION

However, based on the search results you provided, it seems that you are looking to connect a soil moisture sensor to an Arduino. You can follow the instructions from the first link you provided to connect a Gravity: Analog Soil Moisture Sensor to an Arduino UNO. Here are the steps:

1. Connect the sensor's VCC and GND pins to the 5V and GND pins of the Arduino, respectively.

2. Connect the sensor's AOUT pin to one of the analog pins (A0) of the Arduino.
3. Upload the provided code to the Arduino using the Arduino IDE.

The code will print the soil moisture percentage to the Serial Monitor every second. You can modify the code to send the data over a serial connection to an Arduino modem.

Regarding the digital 4-in-1 soil pH and moisture meter, you may need to consult the manufacturer's documentation or contact their technical support to get the specific wiring instructions and output specifications. They may also provide an Arduino library or code examples to simplify the integration process.

In general, to connect any sensor to an Arduino modem, you will need to:

1. Connect the sensor to the Arduino board.
2. Write or use a code example to read the sensor's data.
3. Send the data over a serial connection to the Arduino modem using the `SoftwareSerial` library or a similar library.
4. Configure the Arduino modem to communicate with the desired device or network.

Please note that the specific steps and code may vary depending on the sensor and modem models.